

Student Use and Policies with AI

Supplementary Materials

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What are we doing and Why?

What:

- Using GenAI tools to
 - Augment student learning
 - Assess student interactions
 - Provide feedback for improvement

Why:

- Responsibility to train students
- Generative AI not going away
- Encourage independent learning
- More efficient as educators

Generative AI policy

- Framework of seven levels of GenAI usage
- Instructors are required to designate a level for their course and/or for individual assignments
- Cadets expected to document their GenAI use similar to other resources, full transcript should be retained
- Policy with levels provided to all faculty and cadets
 - Instructors have access to examples of use and documentation for each level

	Level descriptor	What this looks like in practice:
Level 0	No use of GenAI	Cadets will create their own, original work without the use of GenAI in any manner.
Level 1	Organizational / Explanatory use of GenAI	The use of GenAI for personal efficiency (i.e., summarizing notes / readings, clarifying content) is acceptable; however, cadets will create their own, original work without the use of GenAI for submitted assignments. <i>If later used to support an assignment, this use should be acknowledged in documentation statements.</i>
Level 2	Use of GenAI for brainstorming or idea generation	Cadets may consult GenAI as a tool for initial brainstorming or idea generation, but are expected to create their own, original work without the use of GenAI. <i>Use of GenAI in this manner should be acknowledged as part of documentation statements as described by the instructor.</i>
Level 3	Use of GenAI as a feedback tool on student-generated work	Cadets create their own work independently, then may use GenAI as a tool to get feedback about their work (similar to what an instructor might give, e.g., suggestions for improvement, clarification, alignment with the assignment instructions, or editing tips). Cadets are expected to make their own revisions based on this feedback. <i>The cadet should document how and where GenAI was used in their process, potentially including prompts and full transcripts as instructed.</i>

Level 4	Use of GenAI to co-create and/or revise work	Cadets may use GenAI to develop drafts, outlines, or provide suggested revisions to their drafts, but they must critically evaluate, revise, and take ownership of the content. Submitted work should reflect the cadet's understanding, not unedited AI output. <i>This use of GenAI must be clearly documented as instructed, specifying how GenAI contributed and what the cadet did to revise or evaluate that content.</i>
Level 5	Unrestricted, attributed use of GenAI	Cadets may submit work that includes unedited or heavily AI-generated content. The emphasis is on transparency, not authorship. <i>The extent and nature of GenAI use must be clearly disclosed with proper attribution in a documentation statement as described by the instructor.</i>
Level 6	Unrestricted, unattributed use of GenAI	While it is possible to use GenAI in an unrestricted manner without attribution, any who apply this level in their course should carefully consider ethical and legal implications.

* These suggested levels were modified from those adopted at Illinois State University.

Generative AI in my courses

- Augmentation tool for coding, learning, communication
- Learn to write, iterate prompts; question, check responses
- Documentation statements
 - Adapted from USMA
 - Name of the tool, summary of prompts and output, how cadets changed the output/iterated, and transcript link
 - Provide guidance and examples on all assignments
 - Allow revision, with the goal of growth

Documentation statements

1. Name of the tool being used
2. What students provided to ChatGPT (prompts)
3. What students received from ChatGPT (output/transcripts)
4. How they changed the output for the assignment
5. Transcript link

Example: I used ChatGPT 5.2 to help write code for a scatterplot with a regression line. I asked: “How can I plot a scatterplot of height vs weight in R using ggplot2 and add a regression line?” ChatGPT gave me a full ggplot() code snippet. I edited the code to use my dataset name (cadet_data) and customized axis labels and colors.

<https://chatgpt.com/share/6942d3ea-f350-8013-b0a5-55aac51ba119>]

Ethical concerns

- Most LLM platforms are NOT FERPA compliant
 - Redact full student names
- Be aware when uploading copyrighted material
- Check your settings (e.g., ChatGPT paid version)
 - Settings > Personalization > Memory > Manage memories
 - Can turn memory on/off, edit the list of memories that the tool uses
 - Settings > Data controls
 - Can turn data usage on/off

Course descriptions

- Advanced Probability and Statistics
 - Introductory statistics course for majors in the mathematical sciences
- Applied statistical modeling
 - Follow-on survey course in statistical/machine learning
 - Points-free, holistic grading
- Data-Driven Statistics with AI
 - Introductory statistics course for ALL majors
 - Points-free, standards-based grading

Instructing on LLMs

- Introduction to LLMs at the start of the semester
- Explain why or why not
- Demonstrate use for students
- Use it regularly
- Show how to iterate – don't just take the first result
- Highlight good and bad uses
- Demonstrate troubleshooting
- Critical thinking – challenge the LLM
- Teach students how to document

Example Uses

- Daily classroom activities (e.g., interactive reading guides)
- Create new practice problems/GRs
- Administer oral board-type GRs
- Develop assignment rubrics and course grading outcomes
- Assist in grading/providing personalized feedback
- Design assessments (e.g., adapt old project to new data set)
- Positive affirmations (probability themed!)
- So much more!!!

NotebookLM Reading

- Basic intro to GenAI tools at the start of the semester
 - Prompt engineering best practices
- Examples of co-creation and well-structured prompts throughout the course
- Students engage with the reading by explaining their understanding of concepts and working on improvement

Lesson 20 - Bootstrap and Resampling

Shared

Create notebook

Sources

+ Add sources

Search the web for new sources

Select all sources

PDF

IMS Section 12.0-12.1....

PDF

Modern Dive Chapter ...

PDF

Modern Dive Chapter ...

Chat

Lesson 20 - Bootstrap and Resampling

3 sources

These sources explain **statistical estimation** by focusing on **confidence intervals** and the **bootstrapping** method. While a **population parameter** represents a true value, researchers often only have access to a single **sample statistic** to estimate it. To understand how much this estimate might vary, **bootstrapping** uses a computational approach of **resampling with replacement** from the original data to simulate many hypothetical samples. This process allows for the creation of a **distribution of bootstrap means**, which helps determine the accuracy of the sample statistic.

Start typing...

Studio

Audio...

Video...

Mind...

Report...

Flash...

Quiz

Infogr... BETA

Slide... BETA

Data...

Prompt 2 - Quiz m... 7d ago

Add note

Assigned readings and other sources

Chat window

Studio with summaries and tools

Prompts provided as Notes

Reading Guide

- Prompt 1 - Sampling distribution vs bootstrap distribution

Ask the AI: I will explain the difference between a sampling distribution and a bootstrap distribution: [insert your explanation here]. Can you check my explanation for correctness and completeness, and suggest ways to improve my understanding?

- Prompt 3 - Quiz Me (distribution type)

Ask the AI: Can you quiz me by giving me 3-5 short descriptions of a distribution and I'll try to identify whether it's a sample distribution, a sampling distribution, or a bootstrap distribution? After I respond, check if I'm correct and explain any answers I get incorrect.

Prompt Engineering – Best Practices

Remember, it's best to structure your prompt with the following elements:

- Role - who should your tool act as?
- Task - what should your tool do?
- Output - what should the tool's response look like?
- Context - what else does your tool need to know?

PRO TIP: Ask the tool what else it needs to know BEFORE answering

Example Prompts

- Help generating a plot in R

Basic Prompt: Generate a boxplot in R.

Better Prompt: Generate a boxplot in R using the ggplot2 package.

Even Better Prompt: Generate a boxplot in R using the ggplot2 package. Below is an example from my assigned reading: [insert code example from ModernDive].

Example Prompts

- Generating practice problems when studying for the GR

Even Better Prompt: Generate 3-5 practice problems about [insert Block 1/2 topic here (e.g., sampling methodologies, data visualization, numerical summaries)] based on the attached reading assignment. These practice problems should be similar to those provided below from class: [insert example practice problems from your notetakers]. All practice problems should pertain directly to the attached core competencies 1-5 that I'm being assessed on for the next exam. Please also make the practice problems interesting to me - I like basketball, Disney/Pixar movies and Lego sets.

Co-creation with GenAI

- Example 1 - Student forms a plan and presents it to the GenAI tool

Prompt: I am working on a homework problem analyzing the possum dataset in R using tidyverse. I want to compare skull width between possums from Victoria and those not from Victoria.

I think I need to use `group_by()` and `summarise()` to calculate mean and standard deviation. Can you generate tidyverse code that: groups by where the possum comes from, calculates mean, median, standard deviation, and sample size of skull width, removes missing values, and uses the pipe operator? Please explain each line briefly so I can verify I understand it.

Co-creation with GenAI

- Example 2 - Student uses GenAI to create a generic outline for how to approach the problem, then fills in the details

Prompt: I'm in an introductory statistics/data science course using tidyverse. I have this homework problem: [insert homework problem here]

I need to determine whether skull width differs between the two groups using summary statistics only (no hypothesis tests). We've covered measures of center and measures of spread and we're using the tidyverse for data wrangling. Can you help me create a step-by-step outline for how to approach this problem?

Please do NOT solve it — just give a structured plan I can follow."

Co-creation with GenAI

- Bad Example 1 - Student uses GenAI to do the homework problem for them

Prompt: Using the `possum.csv` dataset, determine whether the skull widths of possums from Victoria are different from those not from Victoria. Justify your answer using summary statistics.

Give me an answer that I can easily copy/paste into a `.qmd` file in Posit Cloud.

GR (Exam) Refinements – Spring 2025

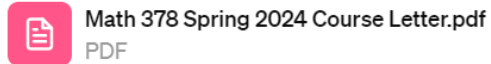
To help you iterate on and improve your understanding of Math 378 course material following the GRs, you have the opportunity to complete a GR Refinement for each block of material.

For this course, you will now utilize an LLM and have a dialogue about what you missed, where you went wrong, and what the correct answer is. You will work with the LLM of your choice to correct your mistakes and refine your understanding of the course material. This will provide you with immediate feedback and allow you to iterate and improve upon your understanding as much as needed. Your full LLM transcript will be submitted to Gradescope as a check of your progress following the GRs and will ultimately be part of the evidence in your semester portfolio, demonstrating your growth regarding the course learning goals.

Previous Work / Examples

Example – Developing a rubric

Hi You



Attached is the course letter for an undergraduate applied statistical modeling course. Can you help me to write descriptions for an A grade, a B grade, a C grade, and a D/F grade in the course? These descriptions should be based on the Learning Goals in the course letter.

PDF Reader

Based on the learning goals outlined in the course letter, here are descriptions for each grade level:

- **A Grade:** Demonstrates exceptional understanding and application of traditional and computationally intensive statistical models, exceeding expectations in advancing computer programming skills. Shows exemplary independent learning habits, with the ability to critically analyze and interpret complex data independently and creatively.
- **B Grade:** Shows strong understanding and application of statistical models, with solid progress in computer programming skills. Exhibits good independent learning habits, with the ability to analyze and interpret data accurately, but may lack some depth or creativity in application.

Hi You

These are a great starting point. Can you please provide more thorough descriptions, touching on each of the three learning goals for each grade?

PDF Reader

Based on the learning goals identified in the course letter for Math 378, here are more detailed grade descriptions:

A Grade

- **Applied Knowledge of Statistical Models:** Demonstrates a superior understanding and application of both traditional and computationally intensive statistical models. Exhibits a high level of skill in utilizing these models to solve complex problems, showing innovation and depth in application.
- **Computer Programming Skills:** Shows exceptional progress in computer programming skills, particularly in R and other relevant software. Able to write efficient, clean, and advanced code, and use programming tools to analyze data and implement statistical models effectively.
- **Independent Learning:** Displays outstanding independent learning habits, consistently going beyond course materials to explore and understand statistical concepts. Demonstrates the ability to learn new tools and techniques autonomously, applying them creatively to course projects and assignments.



Example – Rubric/Grade Descriptions

An “A” grade in Math 378

A student who has earned an A has developed a strong applied knowledge of traditional and computationally intensive statistical models. They thoroughly explore modeling techniques, identify appropriate models, and interpret and evaluate models and output. This student effectively utilizes generative AI to further their understanding of statistical concepts. They thoughtfully craft prompts, actively question responses, and iteratively improve solutions assisted by the AI. This student documents their use of generative models appropriately.

This student has greatly advanced, uses appropriately, and has substantially improved their computer programming skills, particularly in R and/or Python. They understand how to iterate code, debug errors, and how to find and utilize code examples from class, generative AI, and other online resources. They understand how to adjust the code to analyze their own data and can appropriately interpret the output. This student thoughtfully questions the examples to evaluate their appropriateness and iteratively improves solutions.

This student has fully embraced independent and lifelong learning. They take the initiative to explore statistical concepts in more depth. They actively participate in class and collaborate effectively with peers. This student continually improves their statistical knowledge and programming skills over the duration of the course, leveraging resources like their peers and instructor, online learning and generative AI.

LLMs to administer exams

- Used as an interviewer for oral board-type exam
- Customized LLM settings
- Students copied and pasted prompts including the exam question and their answers
- Students consider LLM feedback and iterate/improve on answer

Customized LLM settings

How would you like ChatGPT to respond?

You are an interviewer for an exam in an Applied Statistical Modeling course, essentially administering an oral board over key concepts in the course. I will provide exam questions and my answers to them. At no point should you provide the correct answer to any of the exam questions. You should only consider my answers and then answer the following questions:

Are there any errors in my answer? If so, point them out but do not give the correct answer to the exam question or explain how to fix the errors.

Are there any areas where I need to provide further clarification? If so, point them out but do not give the correct answer to the exam question or explain how to clarify.

I want to use my interactions with you to gauge my comprehension and refine my understanding of the course topics, but I do not want you to give me the correct answer. I will later submit transcripts to my instructor for grading, so I need to show my own understanding and not yours.

Example – exam question and answer

Hi

You

Question 2: The R code presented below implements linear regression on the ``possum`` data set using the `tidymodels` package. Provide brief, informative code comments to explain what each code chunk is doing. Your comments should help explain the code to a fellow analyst who will pick up where your analysis left off in the future.

Answer: The first code chunk creates a `parsnip` model specification for linear regression. The second code chunk fits the linear regression model on the `possum` data set using total length as the response and `pop`, `sex`, head length, and tail length as the predictors. The next code chunk prints a summary of the fitted model, and the `tidy()` function prints the coefficients table with 90% confidence intervals for each model parameter. The `glance()` function prints the model fit statistics. The `predict()` code chunk calculates a 99% prediction interval for a new observation. The `augment()` code chunk calculates predictions for each observation in the `possum` data set. The final code chunk creates diagnostic plots for the fitted model.



ChatGPT

Your comments provide a clear and informative explanation of each code chunk, helping to guide a fellow analyst through the linear regression analysis using the `tidymodels` package. Your descriptions accurately convey the purpose and functionality of each step in the code. However, there is a minor inconsistency in the confidence level mentioned for the confidence intervals generated by the ``tidy()`` function. In the comment, you state a 95% confidence level, but in the actual code, a 90% confidence level is specified. Ensure consistency in your comments to avoid confusion for the future analyst. Overall, your explanation demonstrates a strong understanding of the linear regression workflow in R with `tidymodels`. Well done!

Answer: The slope coefficient for `sexm` is -1.6678. This tells us that for male possums, the average probability decreases by 1.6678. The p-value for the skull width variable is 0.30074. This corresponds to a null hypothesis that the slope for skull width is zero, or that there is no relationship between skull width and population. This value is large, which means that we accept the null hypothesis and conclude that the coefficient is zero.



ChatGPT

Your interpretation of the slope coefficient for ``sexm`` is generally accurate. However, there is a minor error in the explanation of the effect of ``sexm`` on the probability. The coefficient of -1.6678 for ``sexm`` actually represents the log-odds of the probability of a possum being from Victoria (compared to not being from Victoria) for male possums compared to female possums, holding all other variables constant. So, it's more appropriate to say that for male possums, the average log-odds of being from Victoria decreases by 1.6678 compared to female possums.

Regarding the p-value for the ``skull_w`` variable, your explanation is mostly correct. The p-value of 0.30074 suggests that there is no statistically significant relationship between skull width and the probability of a possum being from Victoria. Therefore, we fail to reject the null hypothesis that the coefficient for skull width is zero, implying that skull width is not a significant predictor of the population (``pop``) for a possum in this logistic regression model.

LLMs to assess student interactions

- Customized LLM settings
- Provided exam questions and solutions, rubric
- Asked LLM to provide letter grade and brief, informative feedback to students

Customized LLM settings

How would you like ChatGPT to respond?

I just gave an exam in an Applied Statistical Modeling course, where ChatGPT served as the interviewer, essentially administering an oral board over key concepts in the course.

Students were asked to copy and paste the provided prompts into ChatGPT and provide their answers. They had a chance to receive feedback from ChatGPT and iterate on their initial response, until they felt they had sufficiently demonstrated their understanding on the concept.

ChatGPT was asked to provide feedback, specifically telling the students if there were any errors in their answer and if there were any areas that needed further clarification. ChatGPT was asked not to give the correct answer directly, and not to explain how the answers were incorrect or needing clarification.

I would now like help with grading and providing feedback on these exams. I will provide the exam questions and solutions. I will also provide student answers and a grading rubric. Using the provided solution and grading rubric, give a letter grade for each exam question, and provide brief, informative feedback on the student's response. The feedback should help them toward the correct answer if their answer was incorrect, and provide feedback to help them improve their understanding of the concepts for the future.

Example – grading student interactions

Start a new thread and call it “FirstName Section”.

In the following prompts, I will provide a ChatGPT transcript with a student’s answers to the exam questions. Using the attached grade descriptions and the exam questions and solutions, provide a letter grade and brief, informative feedback to the student on their knowledge of the content, aimed toward helping them to improve their understanding of the concepts in the future. Also provide feedback on their ability to iterate and improve on their answers. Feedback should be no more than a few sentences. Please provide the letter grade and feedback for each of the five exam questions.

Attached is the student transcript. Please provide the feedback based on the previous instructions. The student answers are provided under “Anonymous”, and ChatGPT's response is under “ChatGPT”.

Example – grading student interactions



ChatGPT

Based on the transcript and the guidelines provided in the grading rubrics and exam solutions, here are the letter grades and feedback for each exam question:

Question 1:

- **Grade: B**
- **Feedback:** Your initial response had a major inconsistency with the question's direction on model flexibility. However, your iterative corrections improved the answer significantly, especially in clarifying the relationship between model flexibility and training/test error dynamics, as well as bias and variance. Keep in mind the importance of addressing all parts of the question, including irreducible error, and ensure consistency in terms regarding model flexibility.

Question 2:

- **Grade: A-**
- **Feedback:** Your code comments provided a clear and accurate description of each code chunk's purpose within the linear regression analysis using the `tidymodels` package. Future iterations could benefit from more detailed explanations, especially regarding the model's diagnostic outputs and the implications of the coefficients and confidence intervals.

LLMs to provide feedback on portfolios

- Customized LLM settings
- Provided portfolio instructions, rubric
- Asked LLM to provide letter grade and brief, informative feedback to students
 - Mid-term feedback focused on improvement for end-of-semester

Example – Portfolio grading

Start a new thread and call it “FirstName Section Portfolio”.

Using the attached portfolio instructions and grade descriptions (contained in “0 - Math 378 Spring 2024 Portfolio Requirement.pdf”), provide feedback on the student portfolio. Answer the following questions about the student portfolio:

- Did the student list their letter grades for the graded assignments so far in the course?
- What letter grade did the student give themselves?
- Did the student address their performance in each of the three learning goals?
- Did they answer the assessment questions about their strengths and weaknesses and ways to improve in each of the three learning goals?
- Is the student doing anything to go above and beyond what is required and further develop their understanding/skills?

Then, use the attached portfolio instructions and grade descriptions and the student’s listed letter grades to provide a letter grade and brief, informative feedback to the student on their work so far in the course. The feedback should address each of the three learning goals outlined in the portfolio requirements and should help them to improve their understanding of the course concepts for the future and to grow their skills related to the course learning goals.

Example – Portfolio feedback

Cadet X has systematically listed their letter grades for the graded assignments in the course, addressing their performance comprehensively. They've also engaged with each of the three learning goals by reflecting on their development, challenges, and strategies for improvement in statistical models, coding in R, and independent learning. Cadet X recognizes their strengths in hands-on learning and acknowledges the need for enhancing their conceptual understanding and programming skills. Based on their grades and reflective analysis, Cadet X's self-assessment for an 'A' grade aligns with their demonstrated understanding and proactive approach to overcoming challenges. To further improve, Cadet X might focus on deepening their theoretical grasp during conceptual days and continue to refine their coding skills through practice and exploration of different models and techniques.

Example – Portfolio feedback

Feedback Based on Portfolio Instructions and Cadet Y's Performance:

Learning Goal 3: Independent Learning. Cadet Y's initiative in using additional resources like ChatGPT and creating case studies indicates a commitment to independent learning, aligning with an "A" grade characteristic. However, the recognition of needing improvement suggests there is room to fully embrace this habit, potentially placing them between "A" and "B" performance. Enhancing skills in seeking out and critically evaluating resources will foster this independent learning habit.

Overall Grade: A-/B+ Considering the high performance in coursework (A's) and a slightly lower performance in the exam (B), along with the self-acknowledged areas for improvement, a grade of A-/B+ seems appropriate. This grade acknowledges Cadet Y's strong foundational skills and achievements while recognizing areas for growth.

Recommendations for Improvement: Independent Learning. Continue leveraging external resources and self-directed projects. Seeking feedback from peers and instructors on these endeavors can provide new insights and enhance learning.

Lessons learned

- Grading student interactions took longer initially
 - Check for hallucinations
 - Quality check feedback to students
 - Adjust grading as necessary (missing info)
 - Time was saved in generating consistent, but personalized feedback
- Students did not like LLM-administered exams
 - Uncertainty, something new
 - Too much in the course (AI, points-free, project)
 - Plan to survey on anxiety, comfortability before and after